

# XC Gradient

## Cofounder Strategy Sync

Manufacturing Math · MTTR Deep Dive · Industry 4.0 Roadmap

March 25, 2026 · Oriol · Arnau · Adam

# Today's Agenda

01

## The Manufacturing Math

OEE · OT · PPT · Downtime formula

02

## MTTR Deep Dive

MTTA · MTTD · MTTW · ART — real timing data from the floor

03

## Downtime Decomposition

Planned vs Unplanned · Setup Time · Batch A/B model

04

## Industry 3.0 → 4.0

4-step ladder: IT/OT · Data · Proactive · Autonomy

05

## Our Attack Vector

Where XC Gradient cuts in and expands

06

## Next Steps

Gemba Walk · Decfa & Paver · April 4th sync

# The Manufacturing Math

$$\text{OEE} = \text{Availability} \times \text{Performance} \times \text{Quality}$$

## Availability

$$\text{OT} \div (\text{OT} + \text{Downtime})$$

← *We attack this*

## Performance

$$\text{Actual Speed} \div \text{Ideal Speed}$$

*Future scope*

## Quality

$$\text{Good Units} \div \text{Total Units}$$

*Future scope*

$$\text{Operating Time (OT)} = \text{Planned Production Time} - \text{Downtime}$$

$$\text{PPT} = 600 \text{ min shift} - \text{Schedule Loss (breaks, handovers)}$$

$$\text{Downtime} = \text{Unplanned Stops} + \text{Planned Stops (changeovers)}$$

# MTTR Deep Dive

$$\text{MTTR} = \text{MTTA} + \text{MTTD} + \text{MTTW} + \text{ART}$$

## MTTA

Mean Time To Acknowledge

**~0 min**

From alarm ring to technician notification. Phase 2 automates this completely via OPC-UA integration.

*Phase 2: Autoprompting → 0 min*

## MTTD

Mean Time To Diagnose

**~40 min**

"Shadow Knowledge" phase: searching grease-stained manuals, WhatsApp groups, calling senior tech. Real observed: 40–45 min.

*Phase 1: RAG Copilot → ~5 sec*

## MTTW

Mean Time To Wait

**Variable**

Waiting for spare parts, machine cooldown, tooling. Cannot be fully eliminated but can be reduced with predictive maintenance.

*Phase 3: Predictive ordering*

## ART

Active Repair Time

**~20 min**

Actual wrench-turning time. Physical work software cannot replace — but we ensure mechanics spend 100% of time here.

*Software cannot fix this (yet)*

*"You pay mechanics for ART — but they spend 50% of their time on MTTD acting as librarians. We eliminate the librarian."*

# Downtime Decomposition

$$\text{Downtime} = \text{Unplanned Stops} + \text{Planned Stops}$$

## UNPLANNED STOPS

Breakdown events — machine fails mid-production

*Model: # Breakdown Events  $\times$  MTTR*

**Batch B formula:**

Planned Steps  $\approx N / \sqrt{2}$  (derived from SOP v1)

Batch A: reduce all possible scenarios to lower bound

*XC Gradient: RAG copilot compresses MTTR*

→ Each breakdown event loses less total time

## PLANNED STOPS (Setup Time)

Machine is supposed to be producing — but is stopped for necessary changeovers. Measured by Setup Time.

**Typical Setup Sequence (real observed, ~40 min):**

- ① Staking out tooling for correct batch (60%=3 items)
- ② Warming up the machine
- ③ Calibrating HMI settings
- ④ Running a test part to verify dimensions
- ⑤ Clearing test part, enabling production

*XC Gradient: SOP v1 queries reduce search time*

→ Setup instructions in 5 sec via WhatsApp/tablet

# Industry 3.0 → 4.0: The 4-Step Ladder

To cross the chasm, a factory must climb four rungs:

1

## IT/OT Convergence

Connect the silos. PLCs (OT) and ERP/MES (IT) must speak the same language via OPC-UA, MQTT. IT knows orders; OT knows machine state. Neither talks to the other today.

*XCG: Edge Sensor: passive packet inspection maps the IT/OT boundary without disrupting PLCs.*

2

## Structured & Contextualized Data

Raw sensor streams are noise. A machine error code means nothing without the SOP, the machine model, and the operator's shift history. Data must carry context.

*XCG: S-QDoRA ingests unstructured PDFs, manuals, logs — makes them semantically queryable from Day 1.*

3

## Reactive → Proactive

Stop waiting for alarms. Use historical MTTR data and sensor trends to predict failures before they happen. Run maintenance at planned stops, not unplanned ones.

*XCG: Phase 2: Auto prompting pre-emptively surfaces alerts; Phase 3 schedules predictive maintenance.*

4

## Autonomy & Dynamic Optimization

The factory becomes a self-optimizing system. The scheduling equation automatically routes production to maximize throughput — solving the NP-Hard Job Shop problem in real time.

*XCG: Phase 3 APS: fitness function → dynamic routing → Pareto-optimal throughput vs cost.*

# XC Gradient Attack Vector

## Phase 1

### Diagnostic Copilot

€499/mo

MTTD: 40 min → 5 sec

- Ingest PDFs, manuals, handwritten logs
- RAG pipeline (S-QDoRA, on-prem)
- WhatsApp / tablet query interface
- NIS2 compliant — zero cloud exposure

## Phase 2

### Autoprompting + Cyber Map

€1,499/mo

MTTA: 5 min → 0 sec

- OPC-UA / MES integration
- Passive IT/OT network mapping
- Continuous OT security monitoring
- Alarm auto-routes to correct technician

## Phase 3

### APS Fitness Function

€3,499/mo

OEE: maximize throughput

- Job Shop Scheduling (NP-Hard)
- Real-time machine availability routing
- Predictive maintenance integration
- Pareto-optimal production planning

# Next Steps — Gemba Walk

Decfa & Paver · Before April 4th cofounder sync

01

ALL

## Measure MTTD

When a machine stops, start a stopwatch. Time from alarm to wrench turning. Document with video if possible.

02

Oriel

## Locate Shadow Knowledge

Find grease-stained notebooks, WhatsApp diagnostic groups, sticky notes on HMIs. Photograph and catalog each source.

03

Adam

## Observe Setup / Changeover

Time the full setup sequence. Map each step to the 5-phase model. Record which steps require searching for instructions.

04

Adam

## IT/OT Edge Mapping

Identify where to plug Edge Sensor (mini-PC / Docker). Confirm no PLC disruption risk. Map VLANs if visible.

05

Amau

## Baseline Data Audit

Collect sample documents: PDF manuals, MES exports, operator logs. Assess RAG ingestion complexity for Phase 1 PoV.

06

ALL

## Document Before/After Baseline

Produce ROI case study inputs: current MTTR, OEE estimate, downtime cost per hour. Required for PoV sales motion.